



Analog IC Design Lab (EE653)

Two-Stage Op-Amp Design and Layout

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Design Specifications and Constraints

The two-stage operational amplifier was designed and implemented using Cadence Virtuoso in the SCL 180 nm CMOS technology. The following specifications and constraints were considered during the design.

- **Technology:** SCL 180 nm CMOS
- **Supply Voltage:** $V_{DD} = 1.8\text{ V}$
- **Allowed Voltage Sources:** Only one supply voltage source (V_{DD})
- **Overdrive Voltage:** $V_{OV} > 150\text{ mV}$ for all transistors
- **Tool:** Cadence Virtuoso (Schematic, Layout, DRC, LVS, PEX, Post-Layout)
- **PDK:** SCL 180 nm

Two-Stage Op-Amp Design

Circuit Architecture

The operational amplifier consists of a Miller-compensated two-stage architecture. The first stage is a five-transistor operational transconductance amplifier (5-T OTA), followed by a common-source (CS) second stage. Miller compensation is employed to improve the frequency stability of the amplifier.

Circuit Schematic

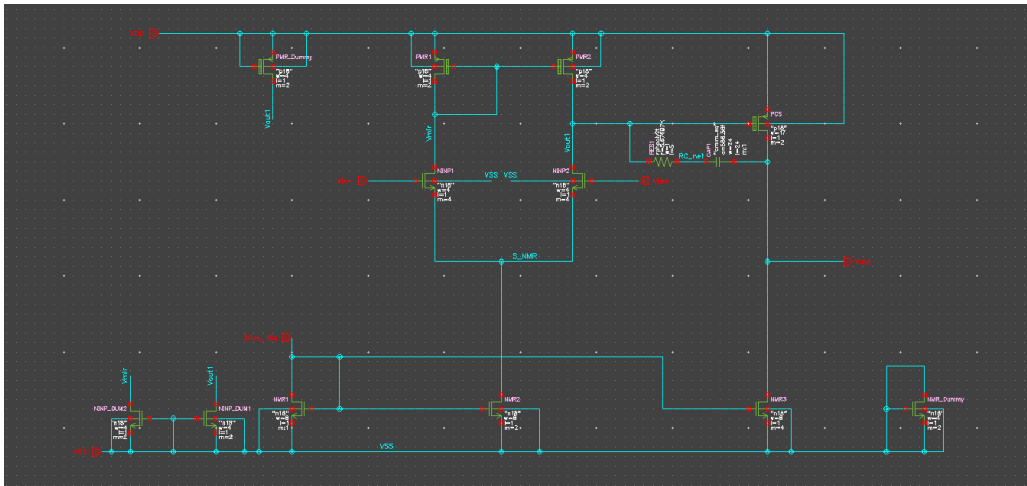


Figure 1: Miller-Compensated Two-Stage Op-Amp Schematic

The first stage provides differential amplification, while the second common-source stage provides additional voltage gain. The compensation network consists of the Miller capacitor and the nulling resistor.

Transistor Sizing

The transistor dimensions and operating currents used in the two-stage Op-Amp are listed in Table 1.

Table 1: MOSFET Dimensions and Drain Currents – Two-Stage Op-Amp

Device	Type	W (μm)	L (μm)	I_D (μA)
Input Pair	NMOS	16	1	10
P Mirror	PMOS	8	1	10
Tail Transistor	NMOS	16	1	20
Bias Transistor	NMOS	8	1	10
M7 (CS Stage)	NMOS	32	1	40
M8 (CS Stage)	PMOS	34	1	40

Passive Component Values

The compensation and load components used in the design are summarized in Table 2.

Table 2: Passive Component Values – Two-Stage Op-Amp

Component	Description	Value
C_C	Miller compensation capacitor	464 fF
R_z	Nulling resistor	7.7 k Ω
C_L	Load capacitor	1 pF

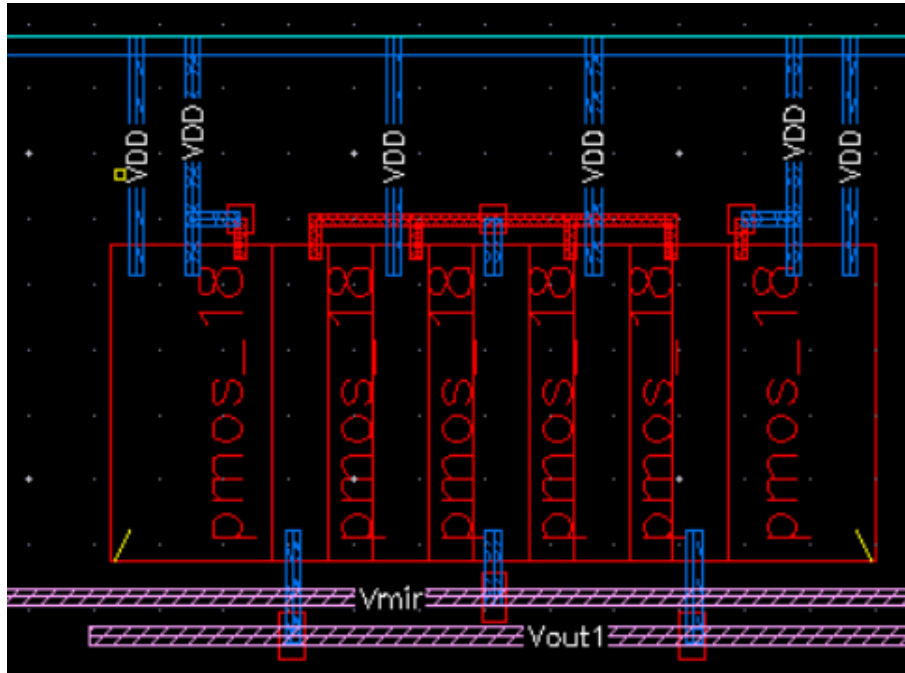
PMOS Current Mirror

Figure 4: PMOS Current Mirror – Common-Centroid Layout

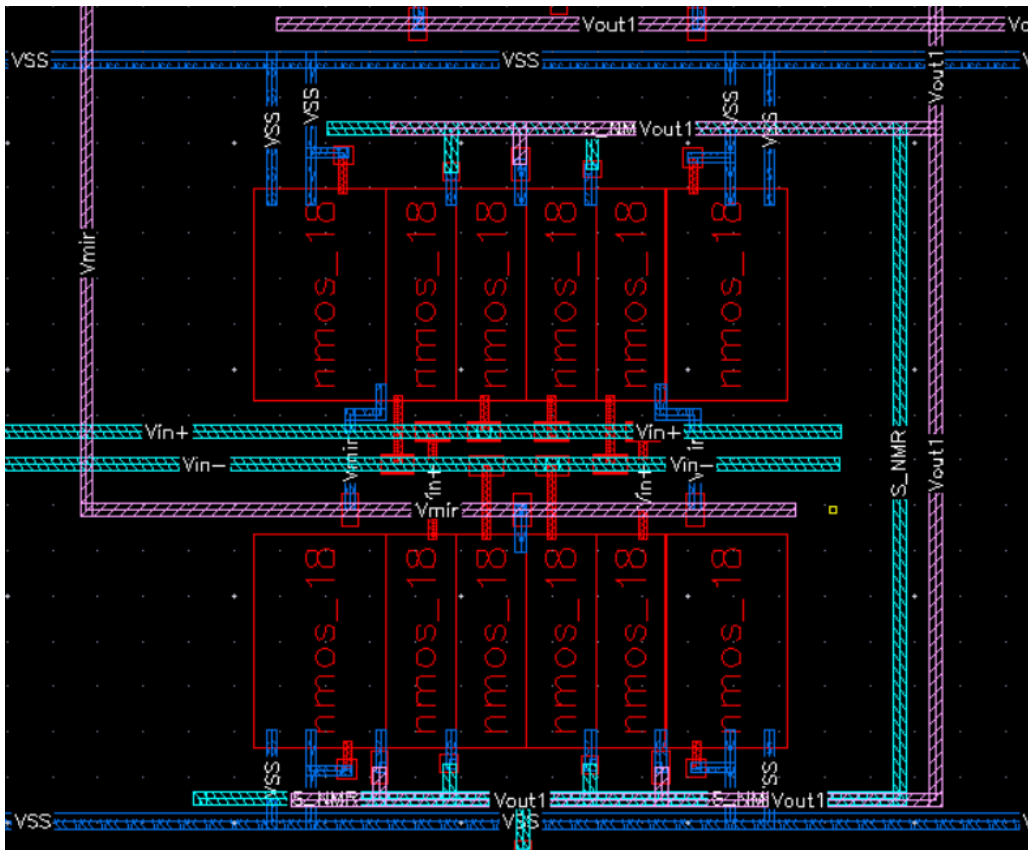
Differential Input Pair

Figure 5: Differential Input Pair – Common-Centroid Layout

Second-Stage and Compensation Layout

Common-Source Stage



Figure 6: Common-Source PMOS Stage Layout

Miller Compensation Network

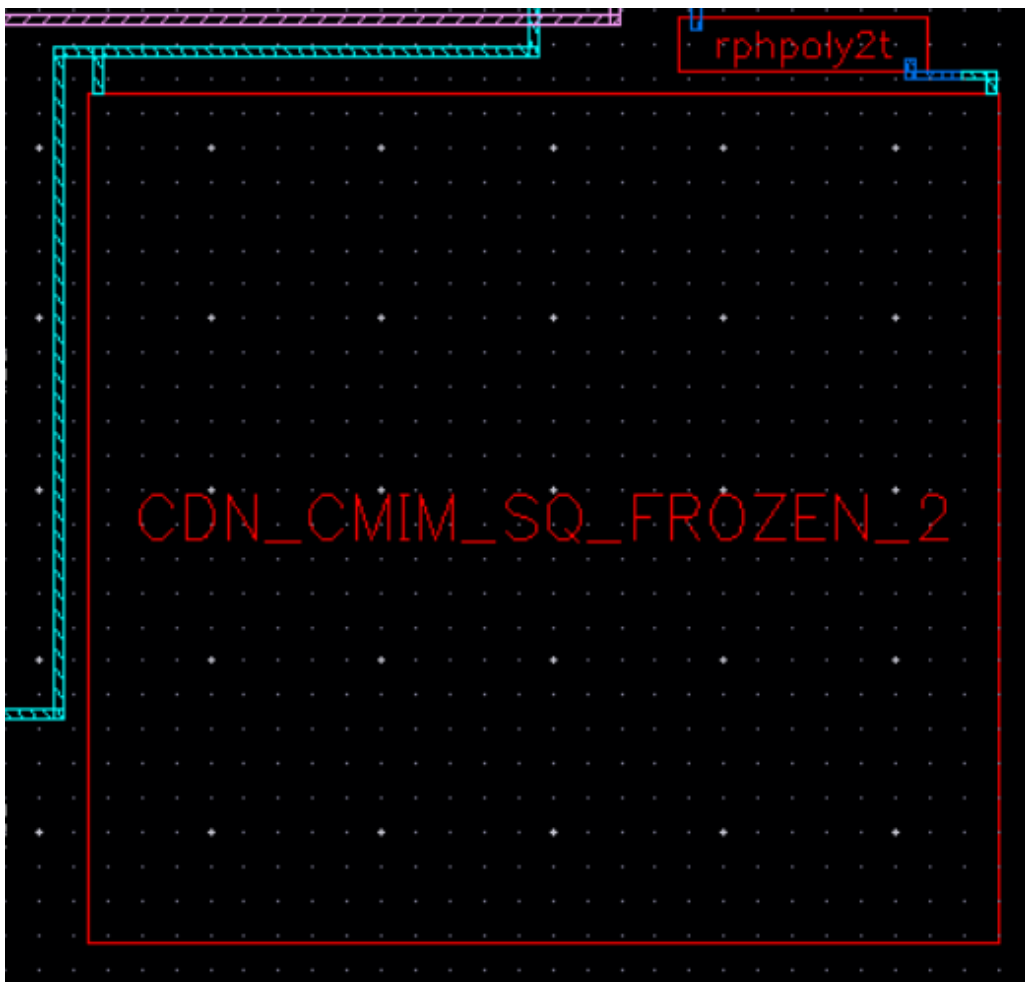
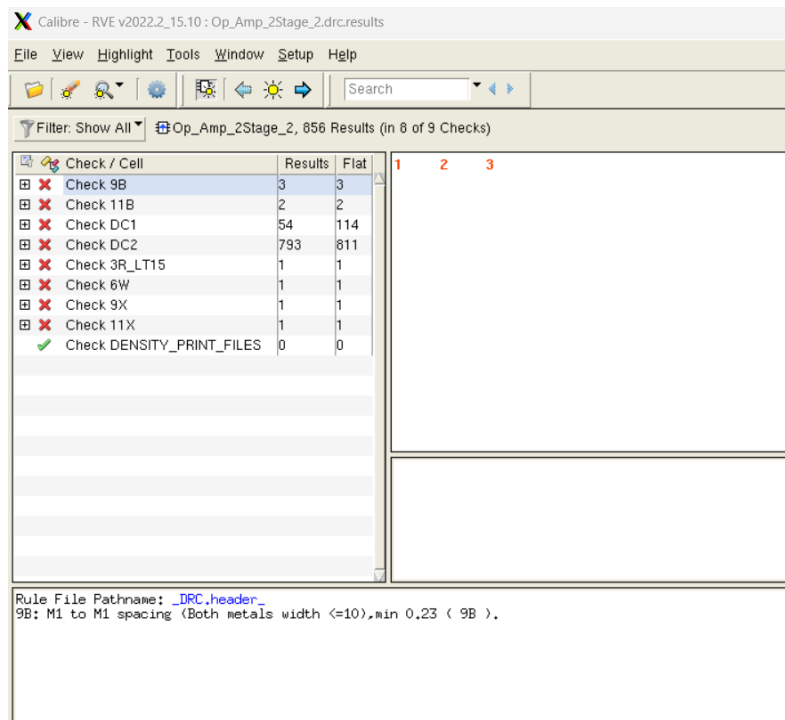


Figure 7: Miller Compensation Capacitor and Nulling Resistor Layout

Physical Verification

The completed layout was verified using Design Rule Check (DRC), Layout versus Schematic (LVS), and Parasitic Extraction (PEX).

Design Rule Check (DRC)



Calibre - RVE v2022.2_15.10 : Op_Amp_2Stage_2.drc.results

File View Highlight Tools Window Setup Help

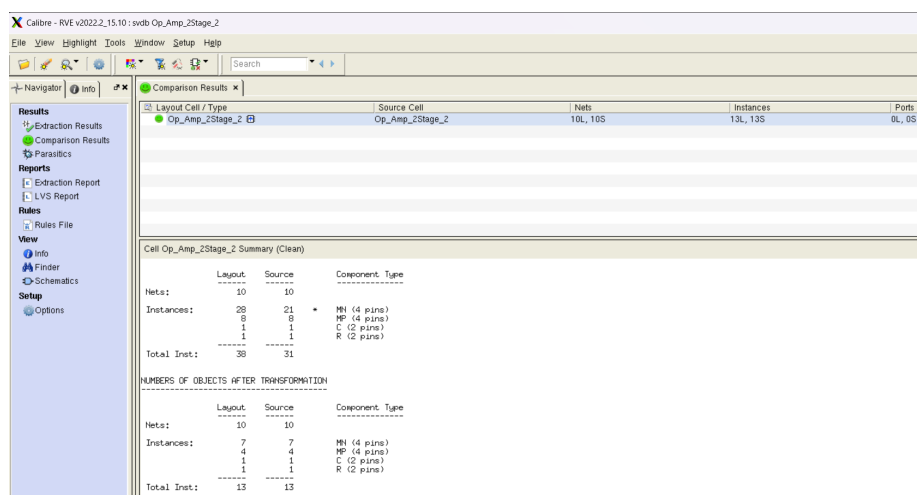
Filter: Show All Op_Amp_2Stage_2, 656 Results (in 8 of 9 Checks)

Check / Cell	Results	Flat
Check 9B	3	3
Check 11B	2	2
Check DC1	54	114
Check DC2	793	811
Check 3R_LT15	1	1
Check 6W	1	1
Check 9X	1	1
Check 11X	1	1
Check DENSITY_PRINT_FILES	0	0

Rule File Pathname: _DRC.header_
 9B: M1 to M1 spacing (Both metals width <=10), min 0.23 (9B).

Figure 8: DRC Results – Two-Stage Op-Amp

Layout Versus Schematic (LVS)



Calibre - RVE v2022.2_15.10 : svdb Op_Amp_2Stage_2

File View Highlight Tools Window Setup Help

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Op_Amp_2Stage_2	Op_Amp_2Stage_2	10L, 16S	13L, 13S	0L, 0S

Cell Op_Amp_2Stage_2 Summary (Clean)

	Layout	Source	Component	Type
Nets:	10	10		
Instances:	28	21	MN (4 pins)	
	8	8	MP (4 pins)	
	1	1	C (2 pins)	
	1	1	R (2 pins)	
Total Inst:	38	31		

NUMBERS OF OBJECTS AFTER TRANSFORMATION

	Layout	Source	Component	Type
Nets:	10	10		
Instances:	7	7	MN (4 pins)	
	4	4	MP (4 pins)	
	1	1	C (2 pins)	
	1	1	R (2 pins)	
Total Inst:	13	13		

Figure 9: LVS Results – Two-Stage Op-Amp

Parasitic Extraction (PEX)

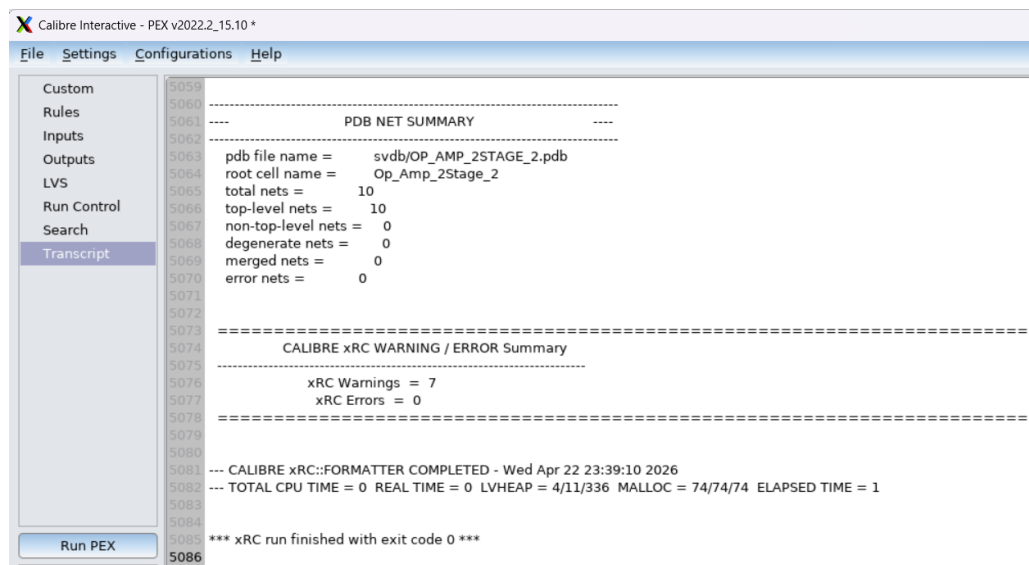


Figure 10: PEX Results – Two-Stage Op-Amp

PEX Extracted View

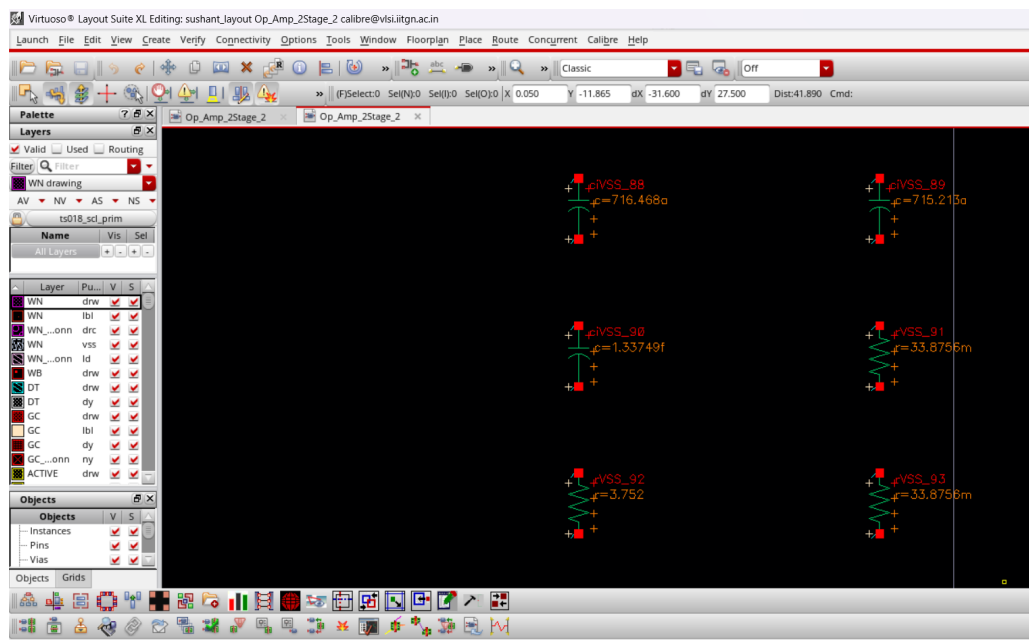


Figure 11: PEX Extracted Calibre View – Two-Stage Op-Amp

Layout Area

The dimensions and total area occupied by the complete two-stage Op-Amp layout are given in Table 3.

Table 3: Layout Area – Two-Stage Op-Amp

Parameter	Value
Width (μm)	36.8
Length (μm)	65.1
Total Area (μm^2)	2396

Pre-Layout and Post-Layout Analysis

The performance of the Op-Amp was evaluated before and after layout. The pre-layout and post-layout results were compared in terms of gain, phase response, transient response, and other key performance metrics.

Frequency Response Comparison

Gain and Phase Comparison

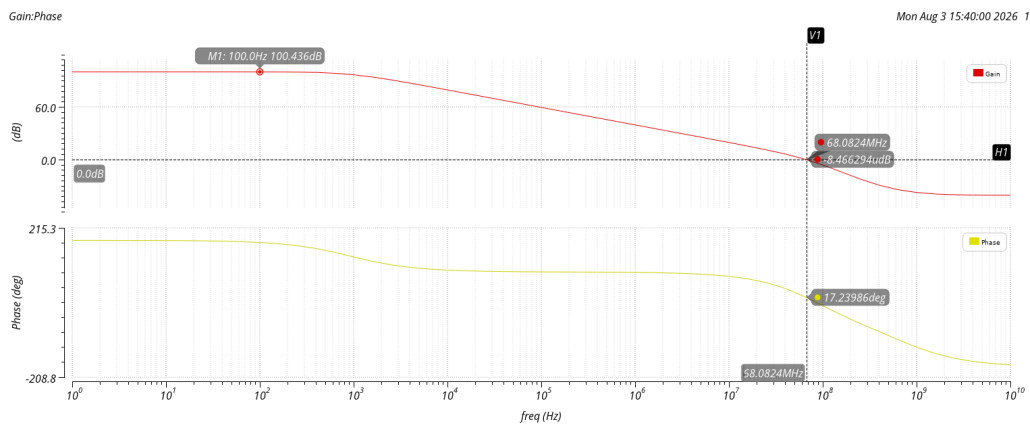


Figure 12: Op-Amp Gain and Phase – Pre-Layout

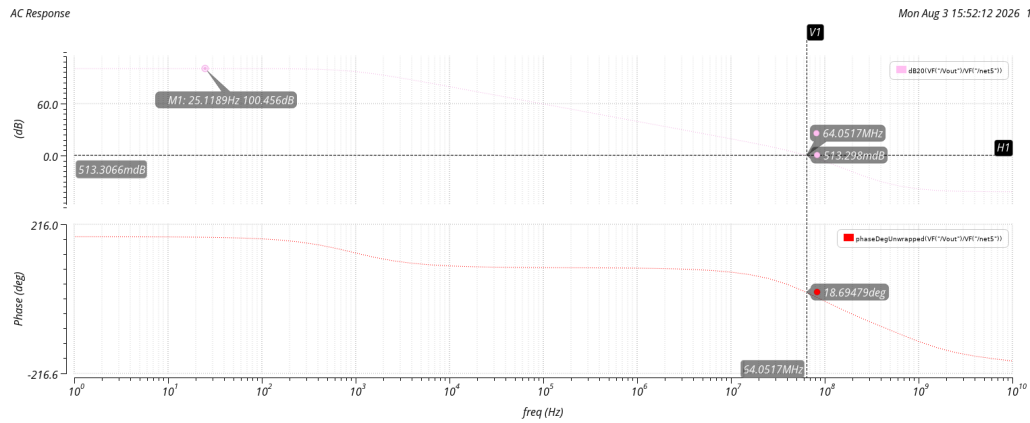


Figure 13: Op-Amp Gain and Phase – Post-Layout

Transient Response Comparison

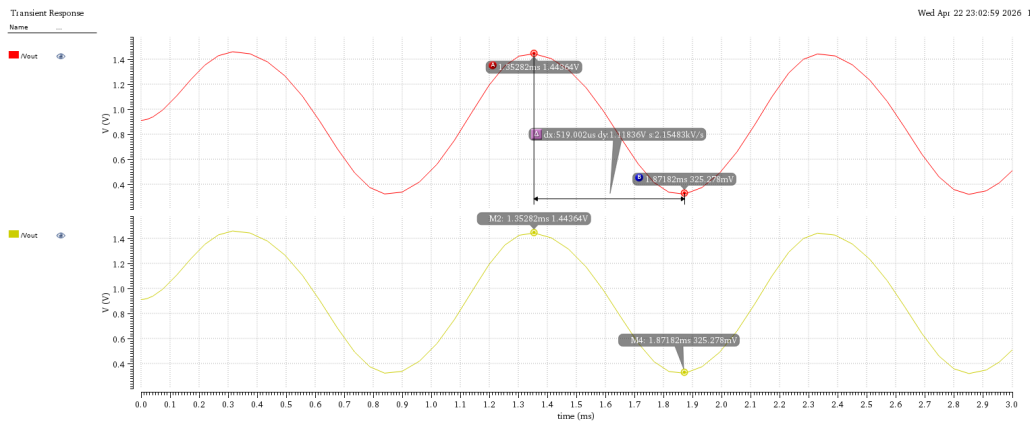


Figure 14: Op-Amp Transient Response Comparison – Pre-Layout vs. Post-Layout

Performance Summary

The key performance parameters obtained from the pre-layout and post-layout simulations are summarized in Table 4.

Table 4: Pre-Layout vs. Post-Layout Performance – Two-Stage Op-Amp

Parameter	Pre-Layout	Post-Layout	Δ (%)
DC Gain (dB)	100.436	100.456	0.02
UGB (MHz)	68.08	64.05	5.88
Phase Margin ($^{\circ}$)	17.23	18.64	1.97
Output Swing (V)	1.14	1.14	0
Power (μ W)	125	125	0

Conclusion

A Miller-compensated two-stage operational amplifier was designed and laid out in SCL 180 nm CMOS technology using Cadence Virtuoso. The design consists of a 5-T OTA as

the first stage and a common-source second stage with Miller compensation.

The physical layout was implemented using appropriate matching techniques, including common-centroid placement for critical transistor pairs and current mirrors. The layout was subsequently verified using DRC, LVS, and PEX. Finally, the pre-layout and post-layout simulation results were compared to evaluate the effect of physical implementation on the Op-Amp performance.